ASIRIVE // CORTEX

IVP Submission //
Theme: User-First or User Fail

ASIRIVE » CORTEX

Navigation
INDEPENDENCE

Made by Haziq, Irfan, Eryna from ADSS

Designed with Singapore Association
Visually Handicapped (SAVH)

https://www.asirive.com

THE ASIRIVE TEAM →

HAZIQ	Systems Architect & Founder. Engineered the dual-track edge-server architecture, YOLO intent-routing, and asynchronous Python pipelines
IRFAN	3d Designer & Validation. Facilitated physical SAVH deployments, live environmental stress-testing, and operational logistics
ERYNA	Hardware Assembly & Data Operations. Managed the $250 BOM unit economics and latency data collection during live SAVH field trials

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THE ASIRIVE TEAM →

IVP MENTOR ⏭

ASST PROF ERKAN KAYACAN

From NTU //

Assistant Professor, School of Mechanical & Aerospace Engineering

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THE ASIRIVE TEAM →

TEACHER MENTOR ⏭

MS NUR SYAZANA BINTE RASHID

MS FLORENCE KUEK

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WHY THIS MATTERS

A visually impaired person must choose between a **white cane for safety** and a **phone for directions** — never both, never hands-free.

Asirive Cortex is a **hands-free wearable guide** that tells you where to go — over GPS outdoors, with the camera indoors, and with live bus arrivals — **without ever touching a phone.**

Navigation independence at 1/10 of the premium price*

***Compared to OrCam My Eye 4**

Navigation → GPS outdoors · camera indoors · live LTA bus arrivals. One voice, zero phone."

Economics → "$186 BOM. 1/10 of the $2,500–3,000 premium."

Validation → "Co-designed with SAVH Singapore — indoors, SAVH campus"

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CONTENTS

01 // Empathize

02 // Define

03 // Market research

04 // Ideate

05 // Prototype

06 // Test

01

EMPATHIZE

RESEARCH THE USER BEFORE DESIGNING ANYTHING

Who are Visually Impaired People, and what does getting around actually feel like

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THE MAJORITY LIVES WHERE ASSITIVE TECH IS LEAST AFFORDABLE

2.2 BILLION PEOPLE Are Visually Impaired

VISION LOSS BY SEVERITY (2020)

*WHO, Blindness & Vision Impairment Fact Sheet (2023); IAPB Vision Atlas (2020).
https://www.who.int/news-room/fact-sheets/detail/blindness-and-visual-impairment

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THE MAJORITY LIVES WHERE ASSITIVE TECH IS LEAST AFFORDABLE

2.2B PEOPLE

WHERE VISION LOSS IS

Most live in Low/Middle Income Countries

Low/middle-income countries — the majority

High-income — the minority

Concentrated in low- & middle-income countries — where $3,000 glasses are out of reach.

*WHO, Blindness & Vision Impairment Fact Sheet (2023); IAPB Vision Atlas (2020).

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WE DID PRIMARY RESEARCH

WITH THE SINGAPORE ASSOCIATION FOR THE VISUALLY HANDICAPPED

We did not just, research online. We went to SAVH for Real User Data

SESSION 1

Online

Met users; learned the cane-vs-phone trade-off.

SESSION 2

Physical @ SAVH

Observed real navigation; tested V1 spatial audio.

SESSION 3

Physical @ SAVH

Tested cane-aware design; gathered feedback.

Result: the cane-aware, navigation-first direction came from users.

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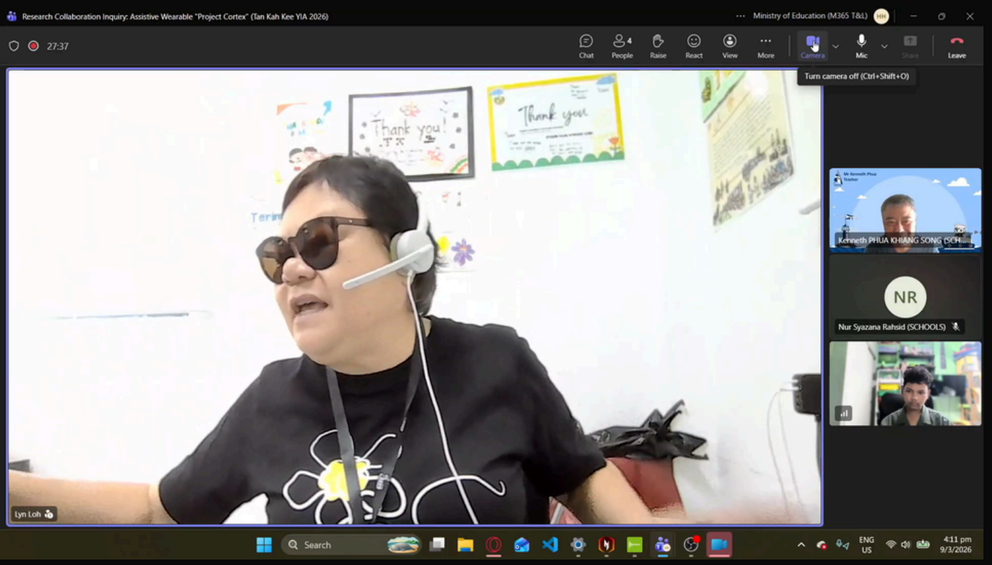
Pictures from our SAVH visit:



>session 1

>session 3

>session 2

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NEVER AT THE SAME TIME

A CANE GIVES SAFETY, A PHONE GIVES DIRECTION

Vision impairment has a significant impact on mobility and independence — yet the tools force a choice between them.

THE CANE

Safe and trusted — but needs your hands and tells you nothing about the route.

THE PHONE

Gives directions — but needs your eyes and hands, away from the ground.

TWO HANDS, TWO TOOLS. THE TRADE-OFF IS THE PROBLEM.

https://www.asirive.com

*WHO — vision impairment significantly impacts mobility & independence (2023); SAVH field observations.

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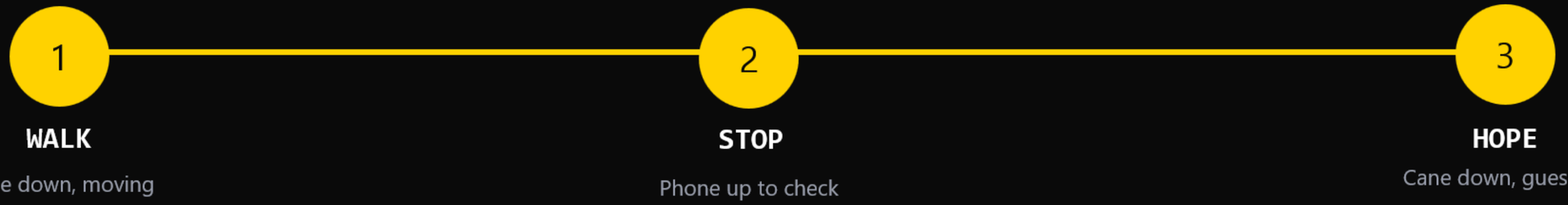
THE STOP AND CHECK

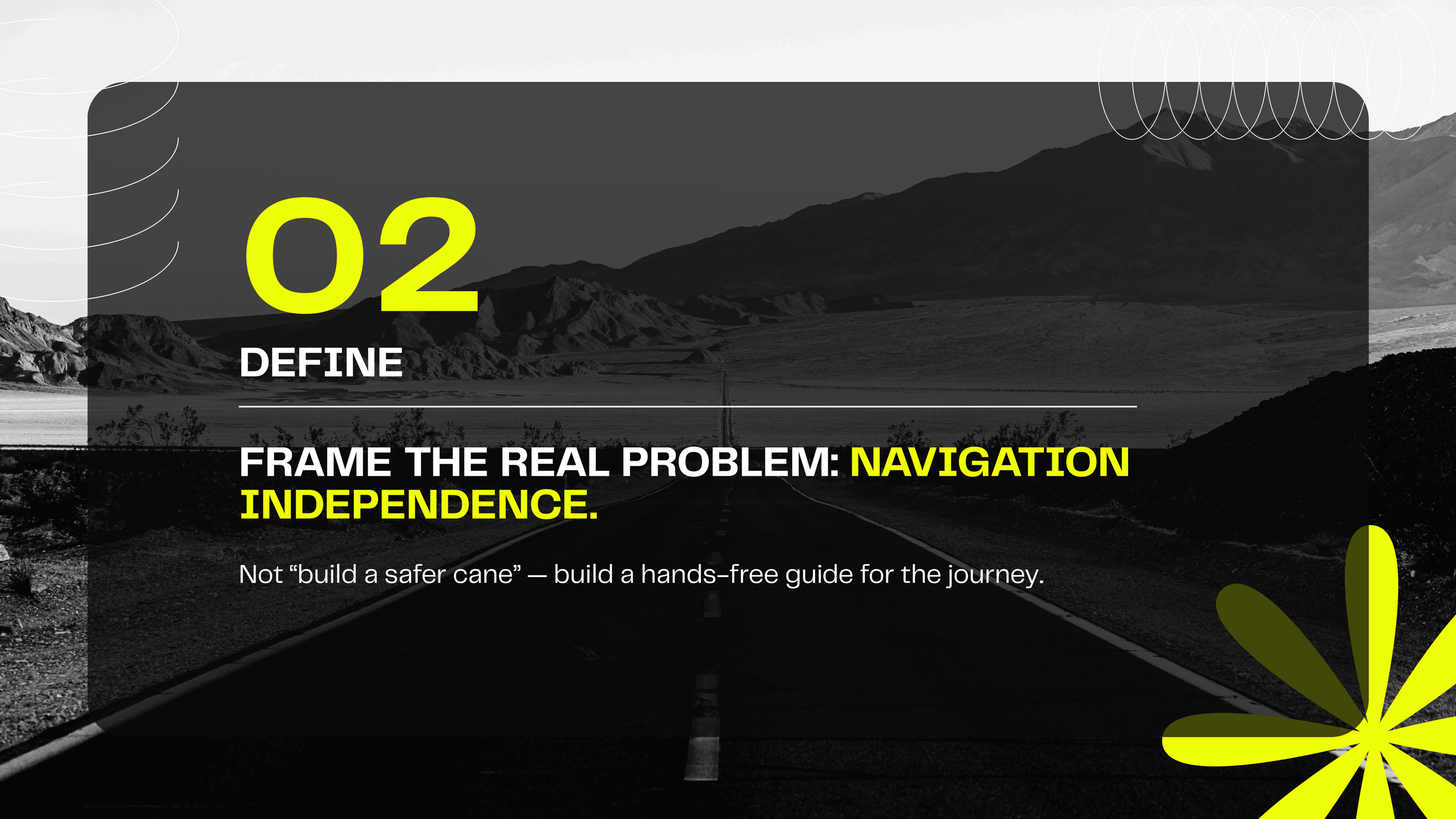
OBSERVED IN THE FIELD

We watched users stop mid-journey to check a phone — then resume, hoping they remembered the next step.



https://www.asirive.com

*Primary research — SAVH field observations (2026): 3 sessions, 1 online + 2 physical.



02

DEFINE

FRAME THE REAL PROBLEM: NAVIGATION INDEPENDENCE.

Not “build a safer cane” — build a hands-free guide for the journey.

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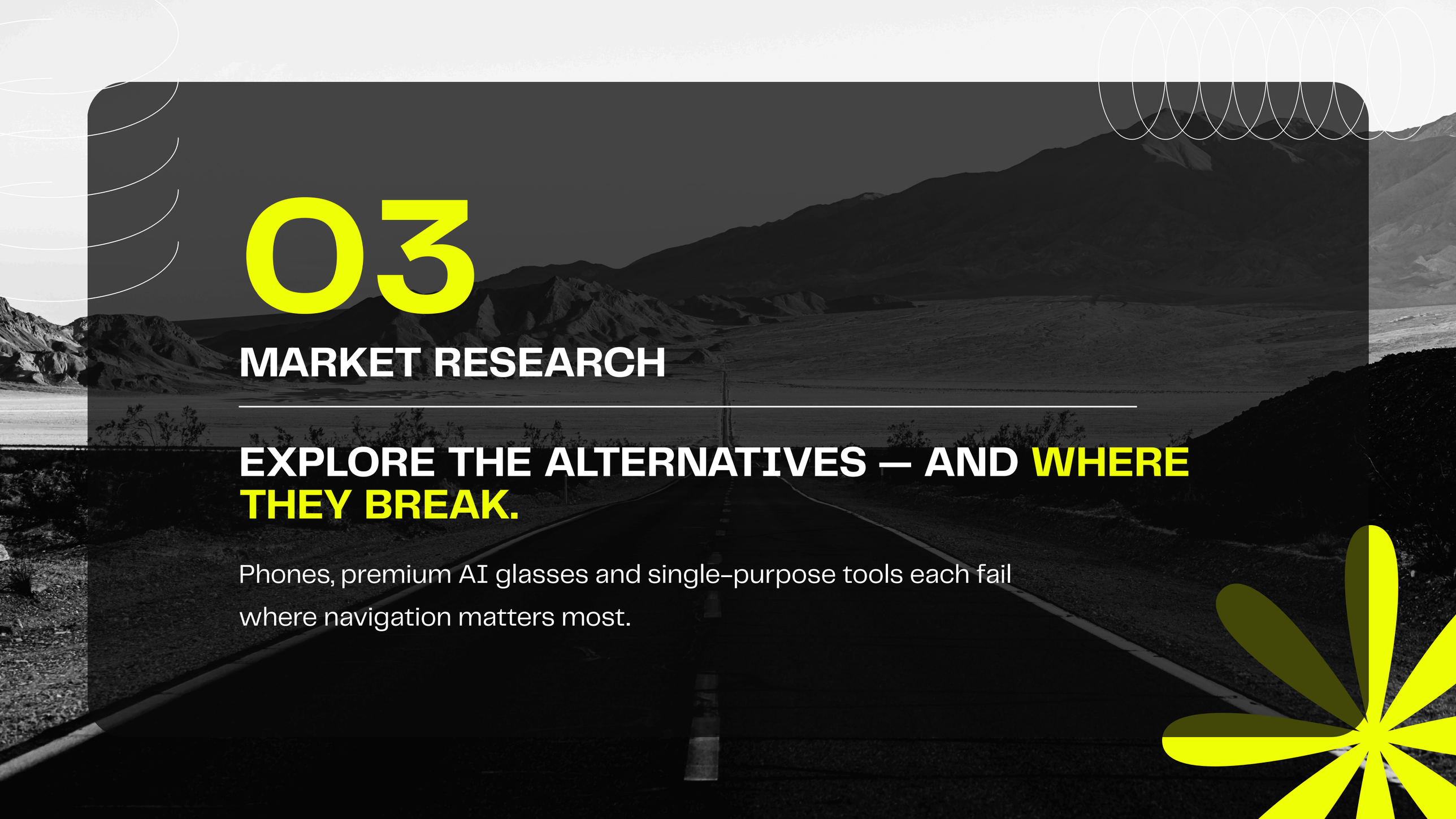
THE PROBLEM FRAMED

NAVIGATION SHOULD NOT CAUSE YOU SAFETY

> How can a visually impaired person get from A to B independently, without putting the cane down, stopping to check a phone, or waiting for a sighted companion?

That is the navigation problem. Everything else is secondary.



03

MARKET RESEARCH

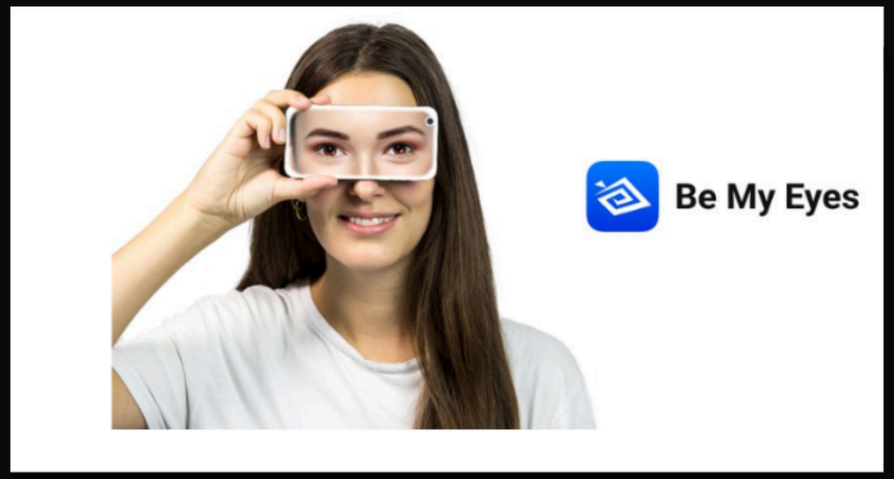
EXPLORE THE ALTERNATIVES — AND WHERE THEY BREAK.

Phones, premium AI glasses and single-purpose tools each fail where navigation matters most.

ALTERNATIVE 1

THE PHONE NEED YOUR HAND

The moment you stop to check the screen, you're blind to the ground. The cane goes in the pocket — and so does your confidence.



‘Be My Eyes’ App

*Primary research — SAVH field observations (2026): 3 sessions, 1 online + 2 physical.

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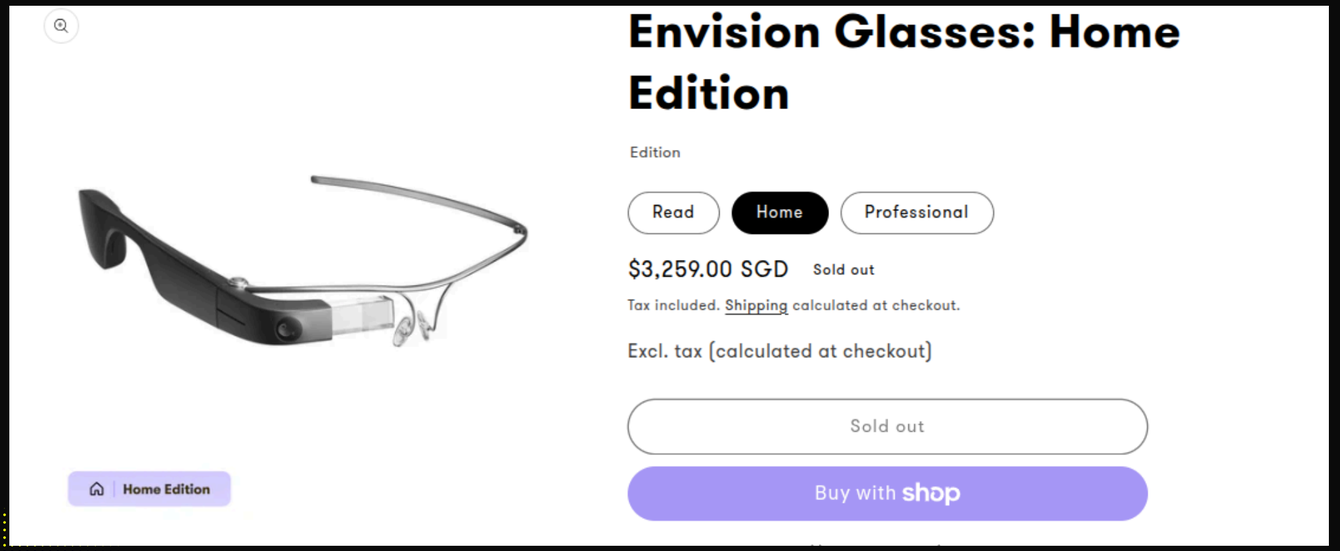
ALTERNATIVE 2

PREMIUM AI GLASSES COST TOO MUCH.

For most of the world's 285 million visually impaired people — and for most of Singapore — that's not an option. Expensive assistive tech stays a luxury.



https://www.asirive.com

https://www.letsenvision.com/glasses/home (link to the shop)

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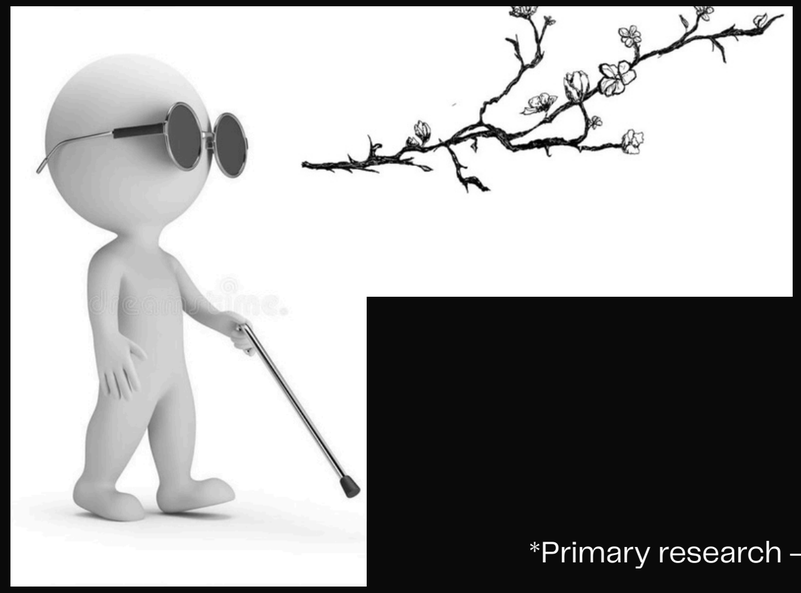
ALTERNATIVE 3

MOST TOOLS COVER THE WRONG HAZARDS.

Many assistants alert on what the cane covers and stay silent on overhead and fast-approaching dangers — the exact blind spots. They add noise, not navigation.

Example of hazard:



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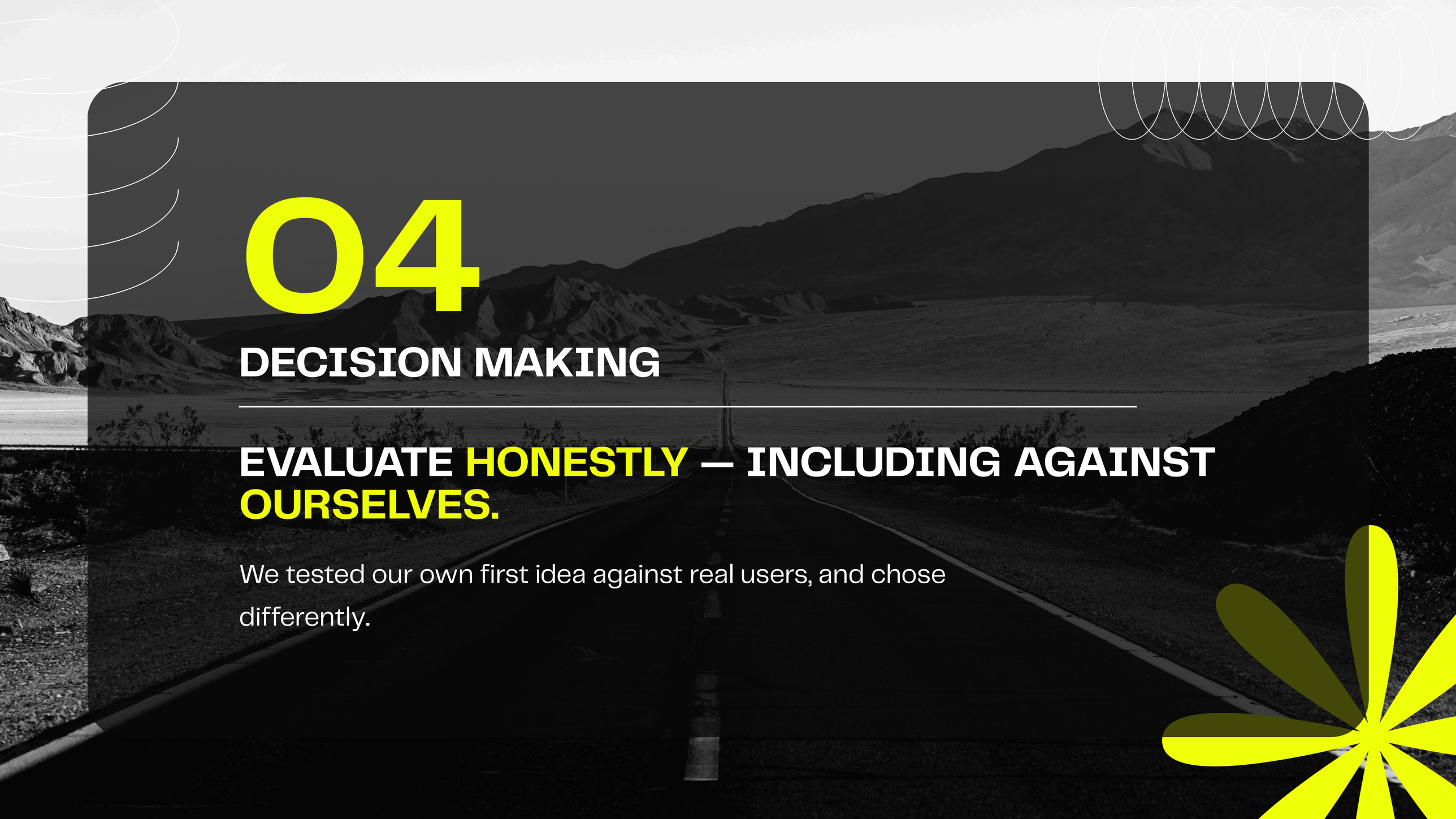

04

DECISION MAKING

EVALUATE HONESTLY — INCLUDING AGAINST OURSELVES.

We tested our own first idea against real users, and chose differently.

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OUR APPROACH

AUDIO-FIRST, CANE-AWARE, NAVIGATION-LED.

A wearable that guides through open-ear audio, respects what the cane already does, and keeps the user's hands and attention free for the journey.

https://www.asirive.com

*Primary research — SAVH field observations (2026): 3 sessions, 1 online + 2 physical.

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COST, INDEPENDENCE DIGNITY

Three tests for every design choice.

COST	Can the people who need it actually afford it?
INDEPENCDE	Does it let them go alone, hands-free?
DIGNITY	Does it respect them — or patronise them?

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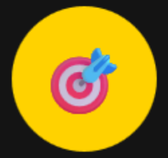
WE CHOOSE NAVIGATION FIRST

NOT “BUILD A SMARTER CANE” —
BUILD A GUIDE FOR THE JOURNEY.

The alternatives all chase safety alerts. The real gap — the thing users actually can't do alone — is navigation. So that's where we put the intelligence.



The real gap

Navigation, not more alerts.

Put intelligence there

Guide the journey.

Safety supports it

As a layer, not the lead.

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OUR FIRST VERSION TRIED TO REPLACE THE CANE.

SAVH TOLD US THE TRUTH — AND WE LISTENED.

Version 1 used full 3D spatial audio for everything. The answer: “Overwhelming. Distracting. We need to hear our surroundings.” So we rebuilt around the cane, not against it.

❌

V1 · Replace the cane

Full 3D spatial audio.

💬

SAVH feedback

“We need to hear the world.”

✅

V2 · Cane-aware

Fill only the blind spots.

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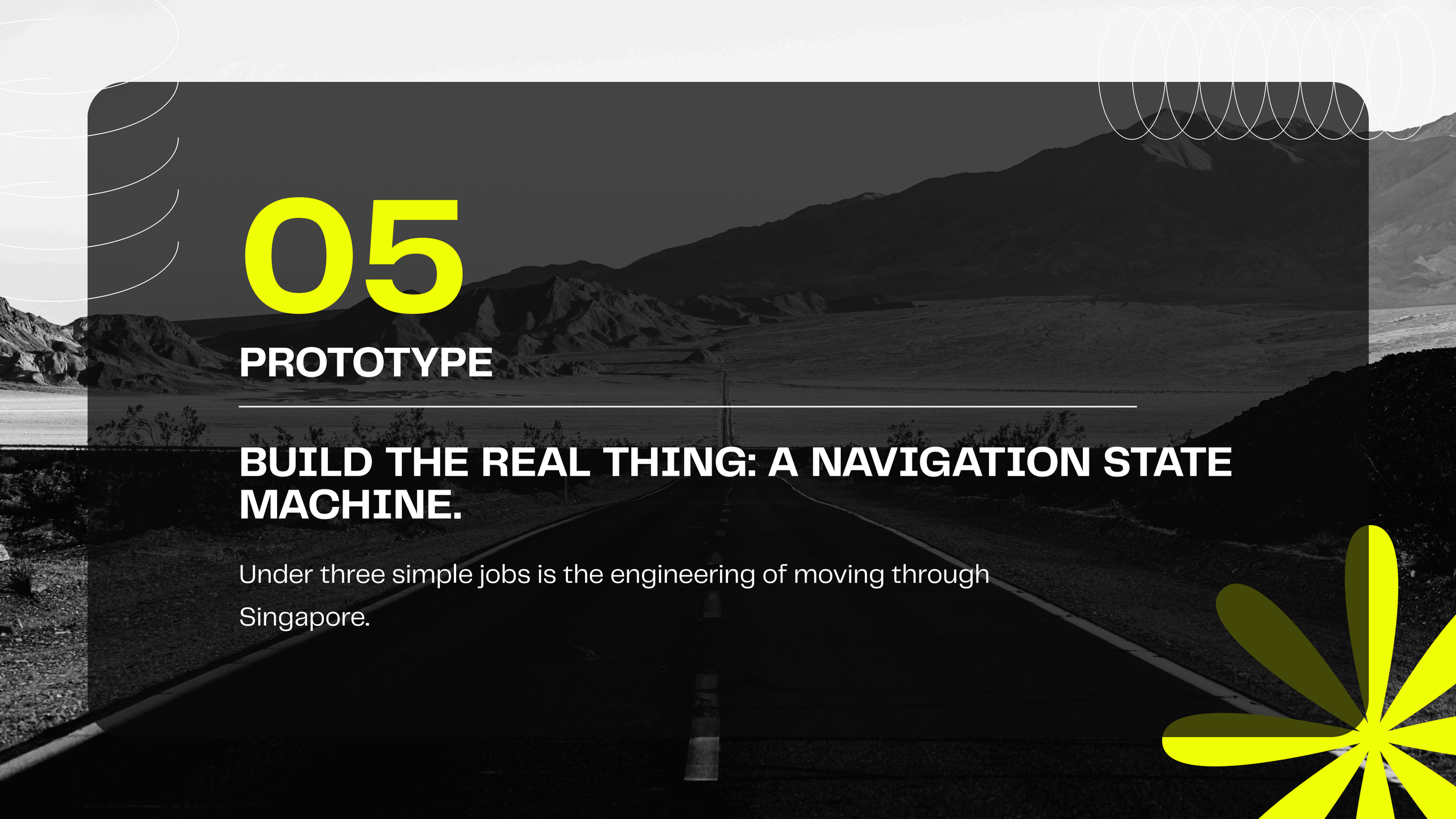
BUILT FOR NAVIGATION. OPEN. 1/10 OF THE PRICE.

CORTEX VS ORCAM

Turn-by-turn navigation	✓	✗
LTA bus arrivals (SG)	✓	✗
Saved places	✓	○
Cane-aware safety	✓	✗
Open architecture	✓	✗
Price	≈$250 retail	$2,500–3,000

https://www.asirive.com

ORCAM READS TEXT. CORTEX GETS YOU THERE.



05

PROTOTYPE

BUILD THE REAL THING: A NAVIGATION STATE MACHINE.

Under three simple jobs is the engineering of moving through Singapore.

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AUDIO-FIRST SMART GLASSES

THE CANE STAYS IN YOUR HAND

The white cane stays your ground sensor; Cortex becomes your guide — through open-ear audio, so you still hear the world around you.

SEES	Reads the space ahead of you.
KNOWS WHERE YOU ARE	GPS, sensors and frequently visited locations
GUIDES YOU WHERE TO GO	Clear spoken directions, hands-free.

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ONE JOURNEY SIX STATES

DOOR-TO-DOOR TRANSIT

“Stay straight for 50 metres. Turn right at the junction.” Spoken in your ear while the cane stays down — directions keep coming as you move.

01 OUTDOOR — GPS turn-by-turn with noise filtering.

02 BUS STOP — Live LTA arrivals + YOLO vision.

03 INDOORS — Camera-led guide where GPS fails.

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HANDS-FREE VOICE-FIRST

THE TRUST LAYER: <100MS SAFETY

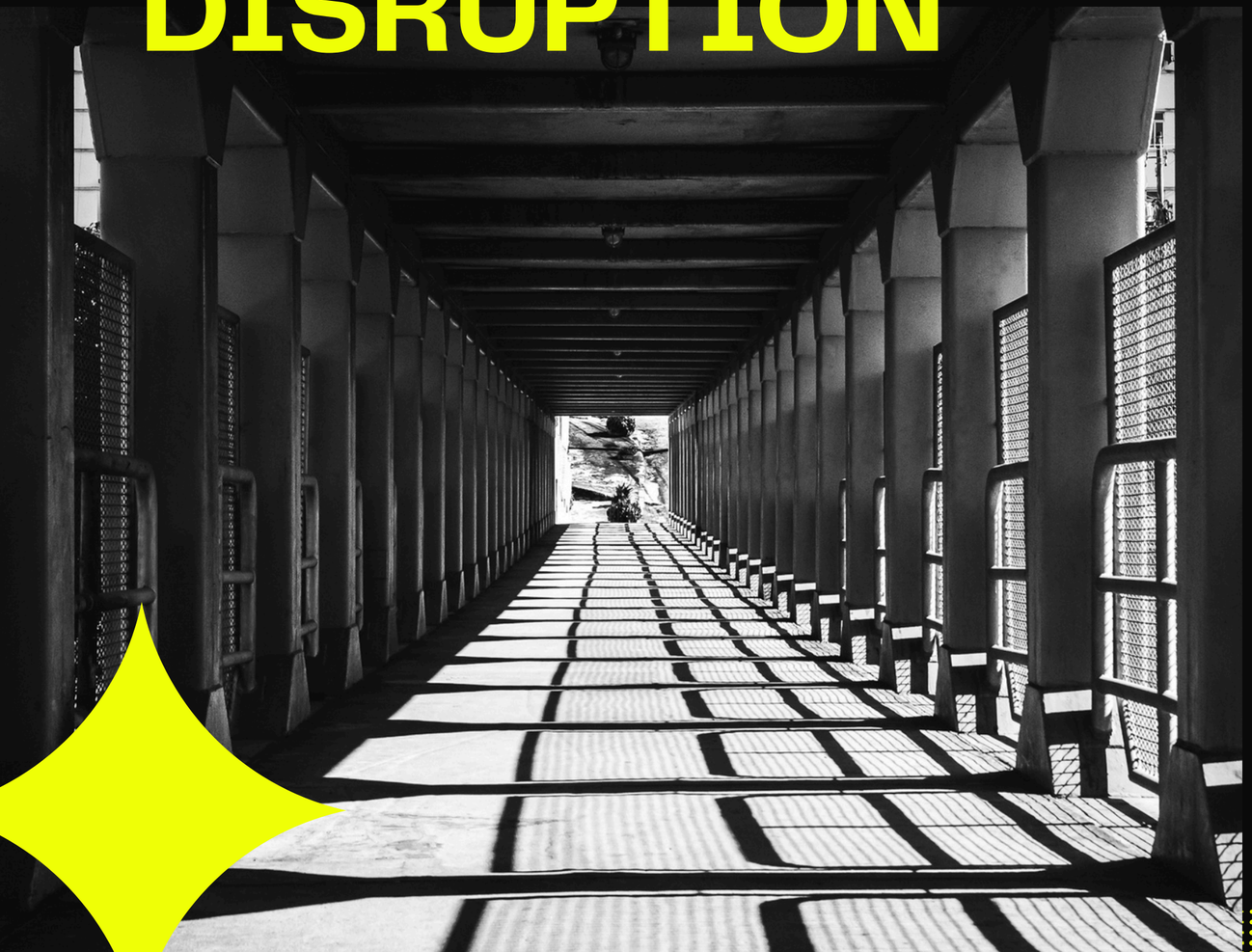
V.2

Cane-aware safety — AI fills the cane-invisible gaps, never duplicates what the cane does.

HAZARD	CANE?	ASIRIVE CORTEX?
GROUND	✓	✗
OVERHEAD	✗	✓ (<R/MS)
STAIRS UP	✗	✓ (<F/MS)
APPROACHES	✗	✓ (<1/0MS)

FEWER FALSE ALERTS. FASTER RESPONSE. LOWER COGNITIVE LOAD.

THE ASIRIVE CORTEX DISRUPTION

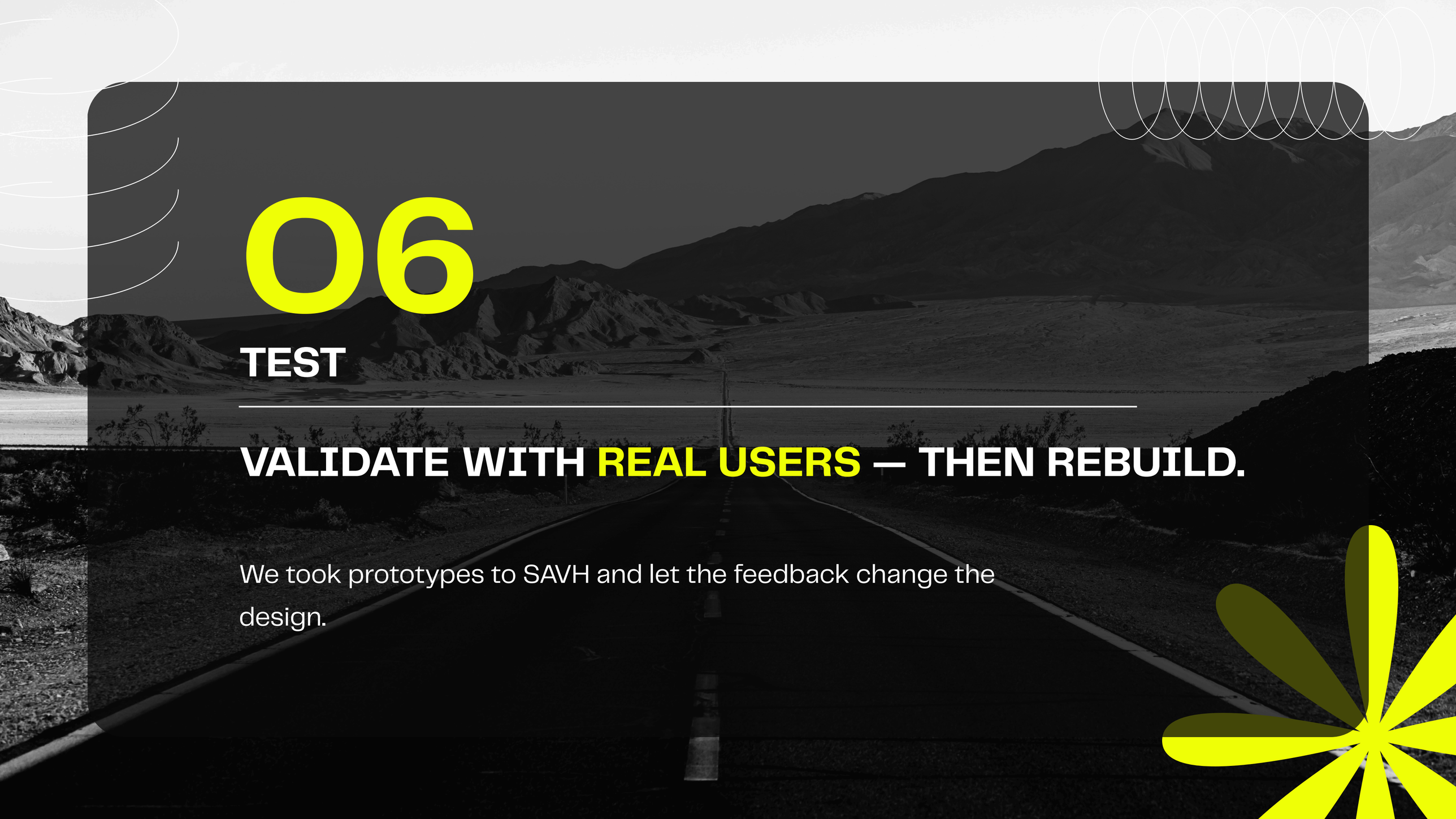


DISRUPTIVE UNIT ECONOMICS

Traditional assistive tech is locked behind a $2,500–$3,000 monopoly. By using commodity edge silicon, Asirive achieves a 90% cost reduction ($186 BOM / $250 retail target), making independence accessible globally.

HYBRID ARCHITECTURAL RESILIENCE

Hazard reflexes (<100ms) and route tracking run locally on-device. Complex scene reasoning streams to Gemini Live. If connectivity drops in MRT tunnels, navigation continues safely offline.

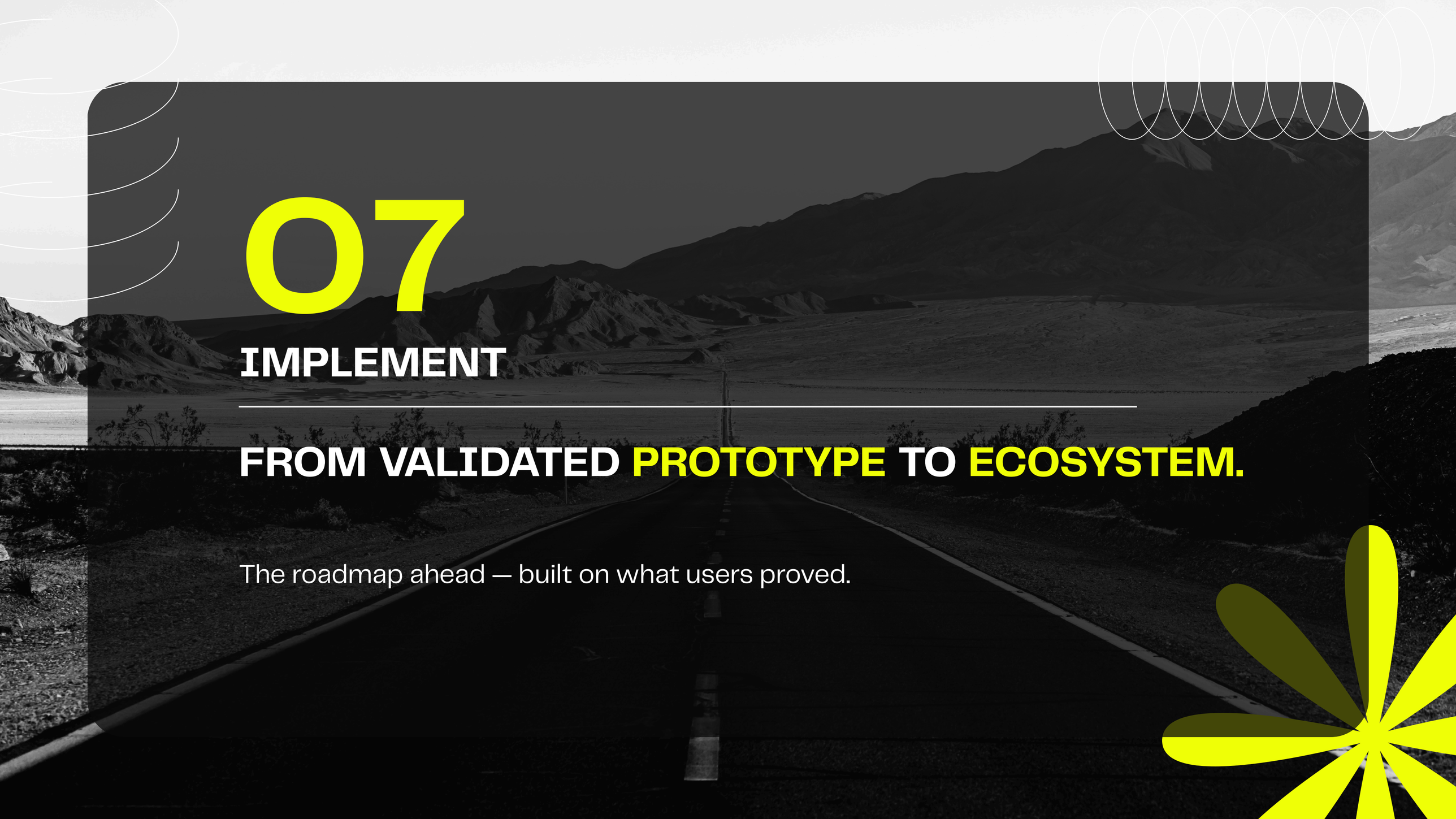


06

TEST

VALIDATE WITH REAL USERS — THEN REBUILD.

We took prototypes to SAVH and let the feedback change the design.



07

IMPLEMENT

FROM VALIDATED PROTOTYPE TO ECOSYSTEM.

The roadmap ahead — built on what users proved.

Breaking the expensive monopoly with hybrid edge-server architecture

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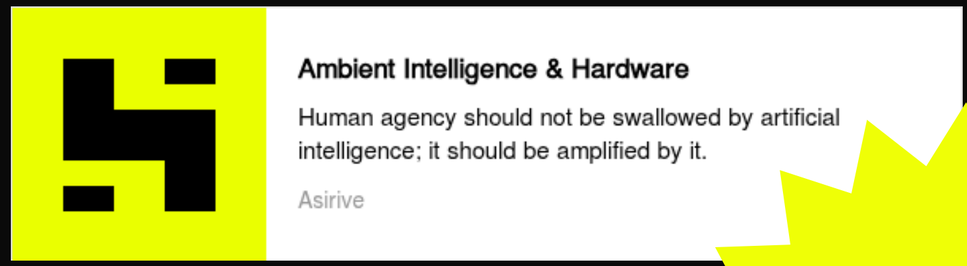
THANK YOU! »»



FOR EXAMPLE

Cane – Covers

Ground – Curbs,

Steps, Holes

Cane – Invisible

Overhang · Stairs

going up ·

Approaching Vehicles

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OTHER OPTIONS DONT WORK EITHER

Phone Trap

App need Hands. Cane goes in the pocket

Price Lock

Premuim AI glasses cost more than most can pay

Too Few Tools

Many Need Help. Few cover the right Hazards

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WE TRIED TO REPLACE THE CANE

SAVH GAVE US FEEDBACK

V.2

Cane-aware safety — AI fills the cane-invisible gaps, never duplicates what the cane does

HAZARD	CANE?	ASIRIVE CORTEX?
GROUND	✓	✗
OVERHEAD	✗	✓
STAIRS UP	✗	✓
OBJECS	✗	✓

FEWER FALSE ALERTS. FASTER RESPONSE. LOWER COGNITIVE LOAD.

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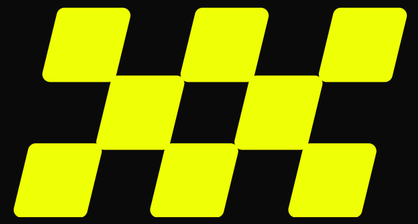
5 SPECIALISTS

1 RULE : SAFETY NEVER LEAVES THE DEVICE



ACTION VOICE

01 GUIDE

Routes intent (97.7% accuracy). Plans walking, bus & MRT legs.

Primary Engine

Always On

02 GUARDIAN

Cane-aware hazard reflex (<100ms). Watches blind spots on Hailo NPU.

Local Edge

Always On

03 THINKER

Powered by Gemini Live. Natural spoken Q&A & scene vision.

Cloud AI

On Demand

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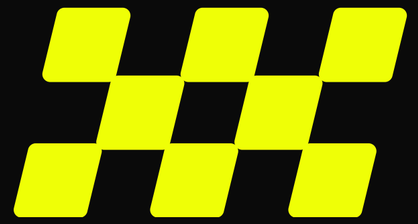


04 LEARNER

Object identification on demand.

'Find my keys' • 'Read sign'.

Edge YOLO

On Demand

05 MEMORY

Remembers fuzzy saved places.

Logs trip context & preferences.

Local SQLite

Passive Sync

COMPONENT	SPEC
• COMPUTE	Pi 5 + Hailo-8L (13 TOPS)
• CAMERA	Camera Module 3 + ToF
• AUDIO	Open-Ear Audio (mic + speaker)
• POWER	USB-C 5000mAh (~8+ hrs)
• WEIGHT	~300g (decoupled load)

$186 BOM // $250 WEARABLE
1/10TH THE COST OF ORCAM ($2,500+)

COMMODITY HARDWARE »

Three Ways Cortex Beats OrCam

Build for Navigation: Turn-by-Turn • LTA Buses • Saved Places

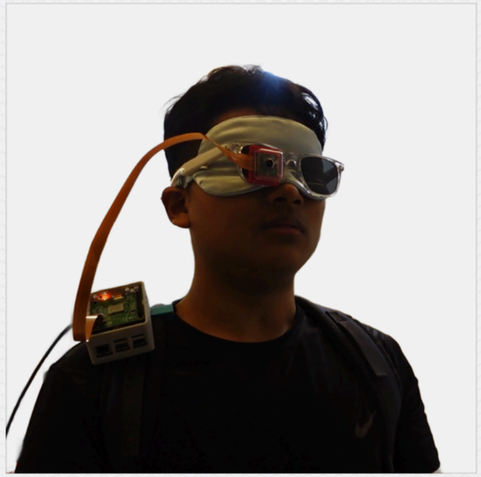
Cane-Aware Safety: <100ms on-device reflex loop

Open Architecture: Commodity hardware stack

OrCam MyEye 3 Pro

~~$4490~~ $4250

VS



Asirive Cortex

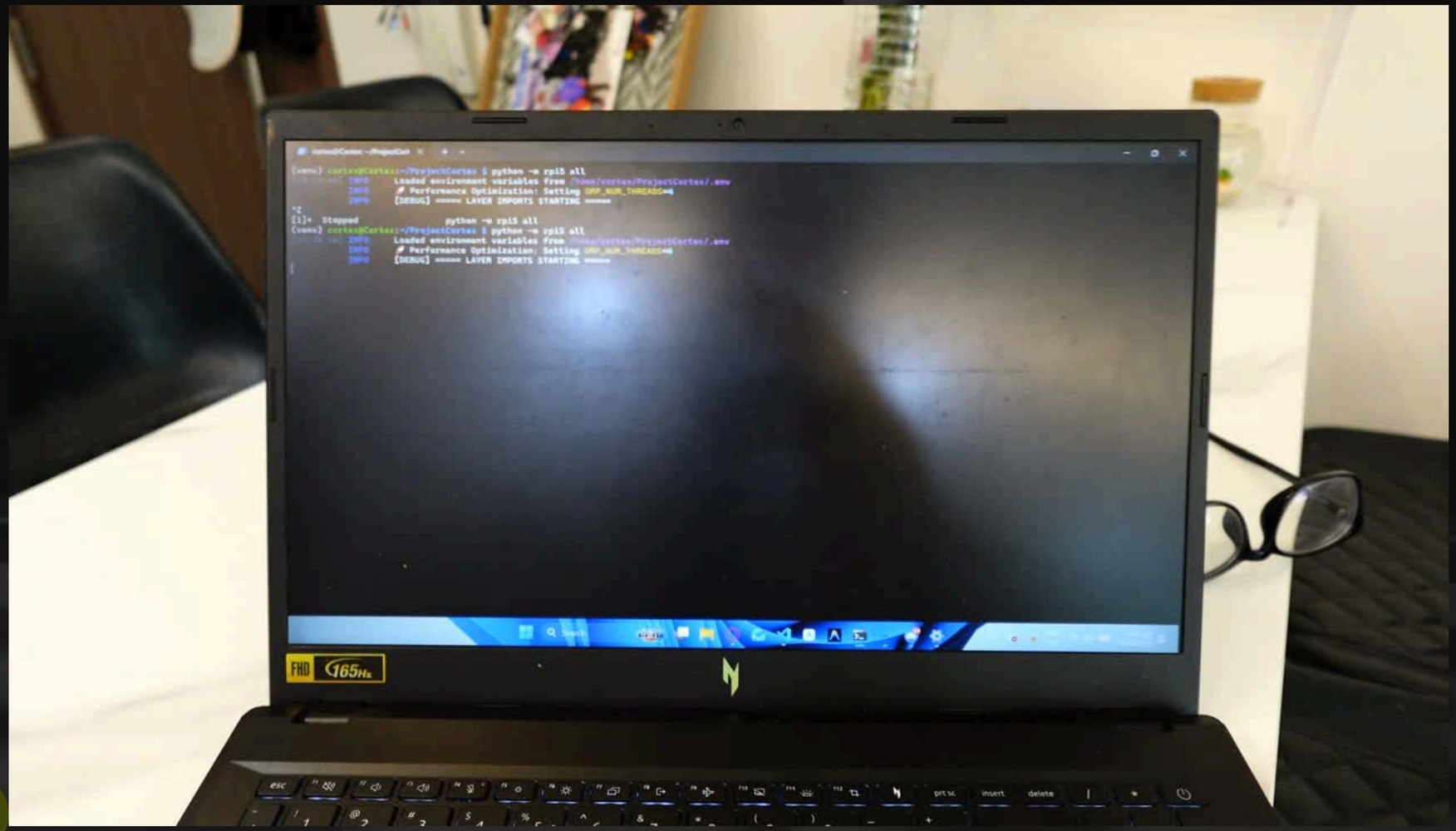
$250 Target Retail

LIVE DEMO

TUI + PROTOTYPE

LIVE DEMO

TUI + PROTOTYPE – VIDEO (SUBMITTED)



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VERSION 1

Monocular Cam

SAVH Validated

Cane Aware Locked

$250

VERSION 2

Binocular Camera + TOF Sensor

More Tests with SAVH

Better Ergonomics

Development of Full Local Navi

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VERSION 3

SAVH Stress Test

Asitive Ecosystem – App

Institutional Pilot

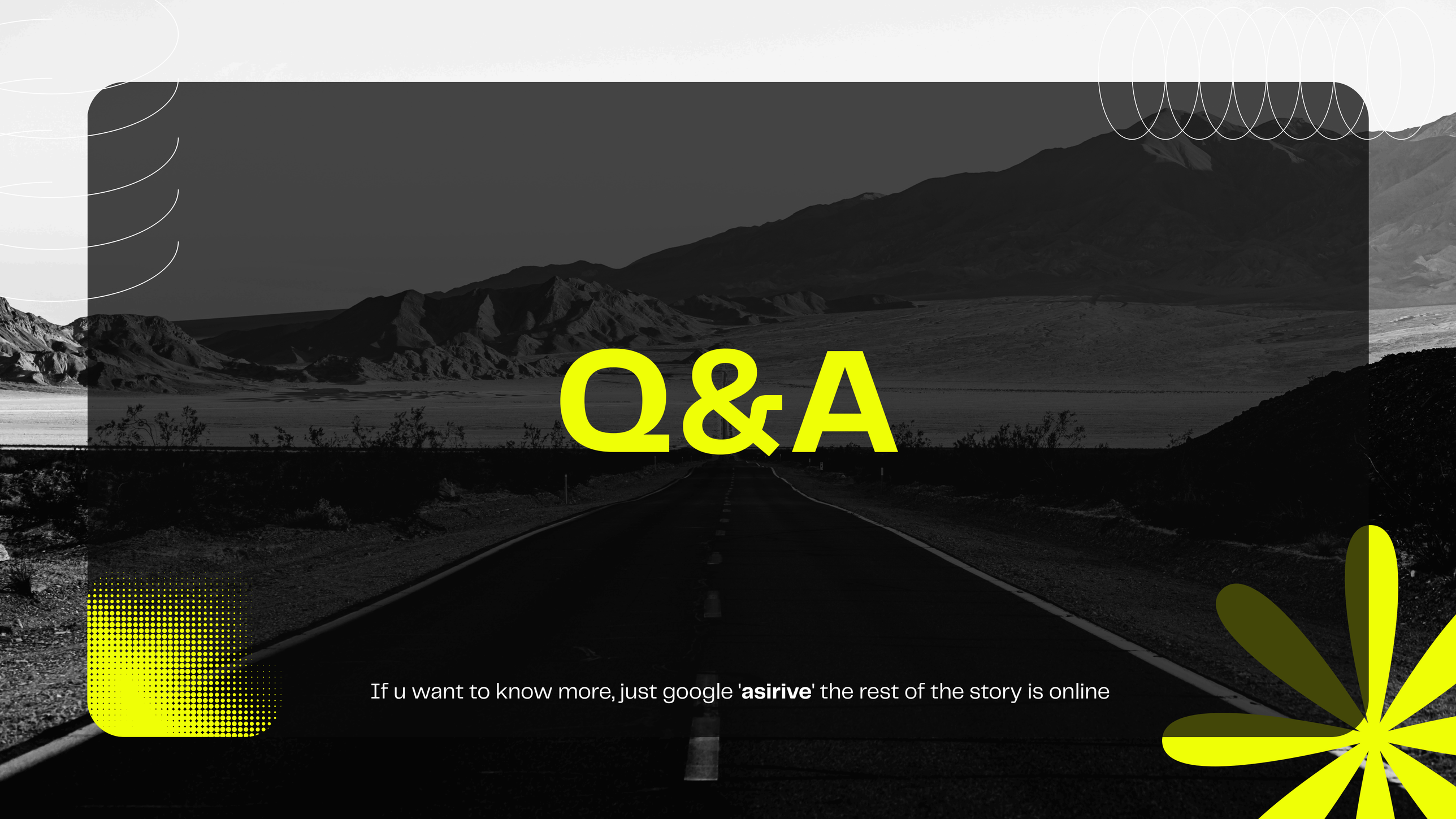
More Units for Testing

Release of Full Local Navi

Improved Ergonomics

YIA WAS A CHECKPOINT

BUT, THE BUILD CONTINUE



Q&A

If u want to know more, just google '**asirive**' the rest of the story is online

Breaking the expensive monopoly with hybrid edge-server architecture

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THANK YOU! »»